Class 6 Notes

Quick revision: Maths, English, Science

Maths

- **Knowing numbers:** Indian and International place value systems. 1 lakh = 100 thousand, 1 crore = 100 lakh. Estimate by rounding. Use brackets to group operations.
- **Factors, multiples, prime numbers:** Prime: only factors 1 and itself. Co-prime: HCF is 1. Divisibility rules: by 2 (last digit even), 3 (digit sum divisible by 3), 4 (last two digits), 5 (ends in 0 or 5), 6 (divisible by 2 and 3), 9 (digit sum divisible by 9), 10 (ends in 0).
- **HCF and LCM:** HCF by prime factorisation or division. For two numbers: HCF × LCM = product of the numbers.
- **Integers:** ..., -3, -2, -1, 0, 1, 2, 3, ... On the number line, numbers increase to the right. Same signs add; different signs subtract. (-3) + (-4) = -7, (-3) + 5 = 2.
- **Fractions:** Proper (3/4), improper (7/4), mixed (1 3/4). Compare by equal denominators. Add and subtract using the LCM of the denominators.
- **Decimals:** Tenths, hundredths. 1.5 = 15/10 = 3/2. Add and subtract by aligning the decimal point. 1 m = 100 cm, so 35 cm = 0.35 m.
- **Ratio and proportion:** Ratio a : b compares two quantities of the same unit. Proportion: a : b :: c : d means a × d = b × c.
- **Algebra:** A variable is a letter standing for a number. Expression: 3x + 5. Equation: 3x + 5 = 20, so x = 5.
- **Basic geometry:** Point, line segment (two ends), ray (one end), line (no ends). Angle types: acute, right, obtuse, straight, reflex.
- **Mensuration:** Rectangle: P = 2(l + b), A = l × b. Square: P = 4a, A = a^2.
- **Data handling:** Tally marks, pictographs, bar graphs. Mean = sum of observations / number of observations.

English

- **Tenses:** Simple, continuous, perfect. 'She is reading' (present continuous), 'She has read' (present perfect), 'She had read' (past perfect).
- **Determiners:** a, an, the, this, that, these, those, some, any, many, much, few, little, every, each.
- **Modals:** can, could, may, might, must, should, would, shall, will. They show ability, permission, possibility or duty. Modals are followed by the base verb: She can swim.
- **Subject-verb agreement:** A singular subject takes a singular verb (He plays) and a plural subject takes a plural verb (They play). Each, every, everyone are singular.
- **Prepositions and conjunctions:** Prepositions link nouns: in, on, at, by, from. Conjunctions join clauses: and, but, because, although, while.
- **Reported speech (introduction):** Direct: He said, 'I am tired.' Indirect: He said that he was tired.
- **Writing:** Informal and formal letters, notice, short story, paragraph. Plan: opening, main points, closing.
- **Vocabulary:** Prefixes: un-, dis-, re-, im-. Suffixes: -ful, -less, -ness. One word for many words: a person who writes books = author.

Science

- **Components of food:** Carbohydrates and fats give energy. Proteins build the body. Vitamins and minerals protect the body. Fibre (roughage) helps digestion. Water is essential.
- **Deficiency diseases:** Vitamin A: night blindness. Vitamin C: scurvy. Vitamin D: rickets. Iodine: goitre. Iron: anaemia.
- **Fibre to fabric:** Cotton and jute are plant fibres. Wool comes from sheep and silk from silkworm cocoons. Spinning makes yarn, weaving or knitting makes fabric.
- **Sorting materials:** Transparent (clear glass), translucent (butter paper), opaque (wood). Soluble (sugar) and insoluble (sand) in water.
- **Separation methods:** Handpicking, winnowing, sieving, sedimentation, decantation, filtration, evaporation (to get salt from salt water).
- **Changes:** Reversible (melting ice, folding paper) and irreversible (burning paper, cooking, rusting).
- **Plants and body movements:** Root, stem, leaf, flower. Joints: ball and socket (shoulder), hinge (knee), pivotal (neck), fixed (skull).
- **Motion and measurement:** Standard unit of length: metre. Types of motion: linear, circular, rotational, periodic.
- **Light, shadows, reflections:** Light travels in a straight line. Luminous objects make their own light (sun). A shadow forms when an opaque object blocks light.
- **Electricity and circuits:** A cell, a bulb, wires and a switch make a circuit. Conductors let current pass (metals); insulators do not (rubber, plastic).
- **Magnets:** Two poles: North and South. Like poles repel, unlike poles attract. Magnetic materials: iron, nickel, cobalt.
- **Air and garbage:** Air is about 78% nitrogen and 21% oxygen. Reduce, reuse and recycle garbage. Compost kitchen waste.